Data Preprocessing



Data Preprocessing

- Why preprocess the data?
- Data cleaning
- Data integration and transformation
- Data reduction
- Discretization and concept hierarchy generation
- Summary

Why Data Preprocessing?

- Data in the real world is dirty
  - incomplete: lacking *attribute values*, lacking certain *attributes of interest*, or containing only aggregate data
  - noisy: containing errors or outliers
  - inconsistent: containing discrepancies in codes or names
- No quality data, no quality mining results!
  - Quality decisions must be based on quality data
  - Data warehouse needs consistent integration of quality data
  - Required for both OLAP and Data Mining!

Why can Data be Incomplete?

- Attributes of interest are not available (e.g., customer information for sales transaction data)
- Data were not considered important at the time of transactions, so they were not recorded!
- Data not recorder because of misunderstanding or malfunctions
- Data may have been recorded and later deleted!
- Missing/unknown values for some data

Why can Data be Noisy/Inconsistent?

- Faulty instruments for data collection
- Human or computer errors
- Errors in data transmission
- Technology limitations (e.g., sensor data come at a faster rate than they can be processed)
- Inconsistencies in naming conventions or data codes (e.g., 2/5/2002 could be 2 May 2002 or 5 Feb 2002)
- Duplicate tuples, which were received twice should also be removed

Multi-Dimensional Measure of Data Quality

- A well-accepted multidimensional view:
  - Accuracy
  - Completeness
  - Consistency
  - Timeliness
  - Believability
  - Value added
  - Interpretability
  - Accessibility
- Broad categories:
  - intrinsic, contextual, representational, and accessibility.

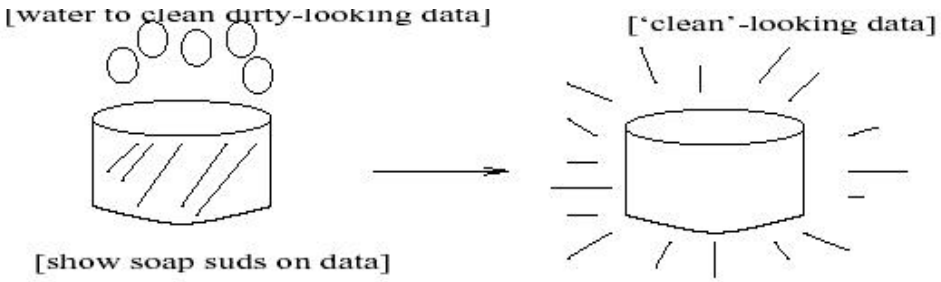
Major Tasks in Data Preprocessing

- Data cleaning
  - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- Data integration
  - Integration of multiple databases, data cubes, or files
- Data transformation
  - Normalization and aggregation
- Data reduction
  - Obtains reduced representation in volume but produces the same or similar analytical results
- Data discretization
  - Part of data reduction but with particular importance, especially for numerical data

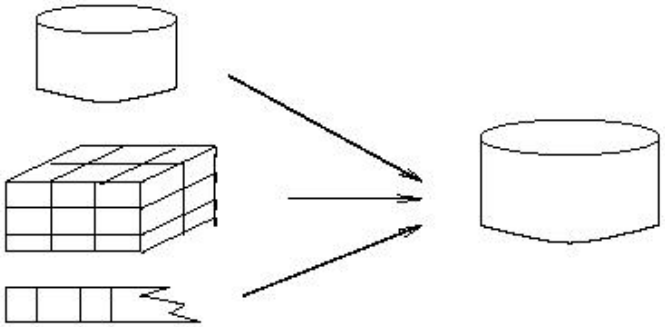
outliers=exceptions!

Forms of data preprocessing

Data Cleaning



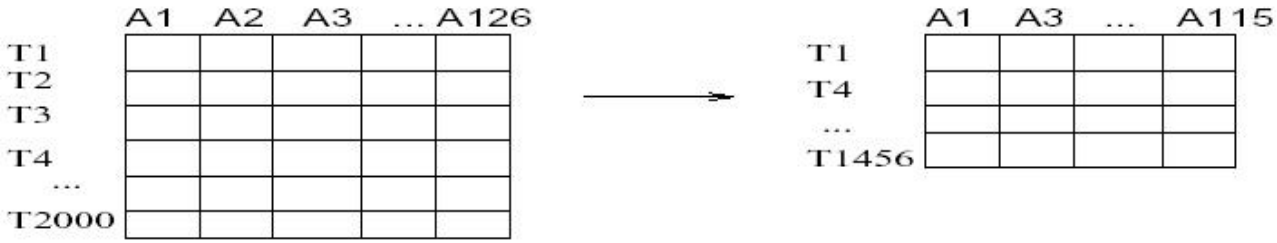
Data Integration



Data Transformation

-2, 32, 100, 59, 48 → -0.02, 0.32, 1.00, 0.59, 0.48

Data Reduction



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Data Cleaning

- Data cleaning tasks
  - Fill in missing values
  - Identify outliers and smooth out noisy data
  - Correct inconsistent data

How to Handle Missing Data?

- Ignore the tuple: usually done when class label is missing (assuming the tasks in classification)—not effective when the percentage of missing values per attribute varies considerably.
- Fill in the missing value manually: tedious + infeasible?
- Use a global constant to fill in the missing value: e.g., "unknown", a new class?!
- Use the attribute mean to fill in the missing value
- Use the attribute mean for all samples belonging to the same class to fill in the missing value: smarter
- Use the most probable value to fill in the missing value: inference-based such as Bayesian formula or decision tree

How to Handle Missing Data?

Age	Income	Religion	Gender
23	24,200	Muslim	M
39	?	Christian	F
45	45,390	?	F

Fill missing values using aggregate functions (e.g., average) or probabilistic estimates on global value distribution

E.g., put the average income here, or put the most probable income based on the fact that the person is 39 years old

E.g., put the most frequent religion here

Noisy Data

- Noise: random error or variance in a measured variable
- Incorrect attribute values may exist due to
  - faulty data collection instruments
  - data entry problems
  - data transmission problems
  - technology limitation
  - inconsistency in naming convention
- Other data problems which requires data cleaning
  - duplicate records
  - incomplete data
  - inconsistent data

How to Handle Noisy Data? Smoothing techniques

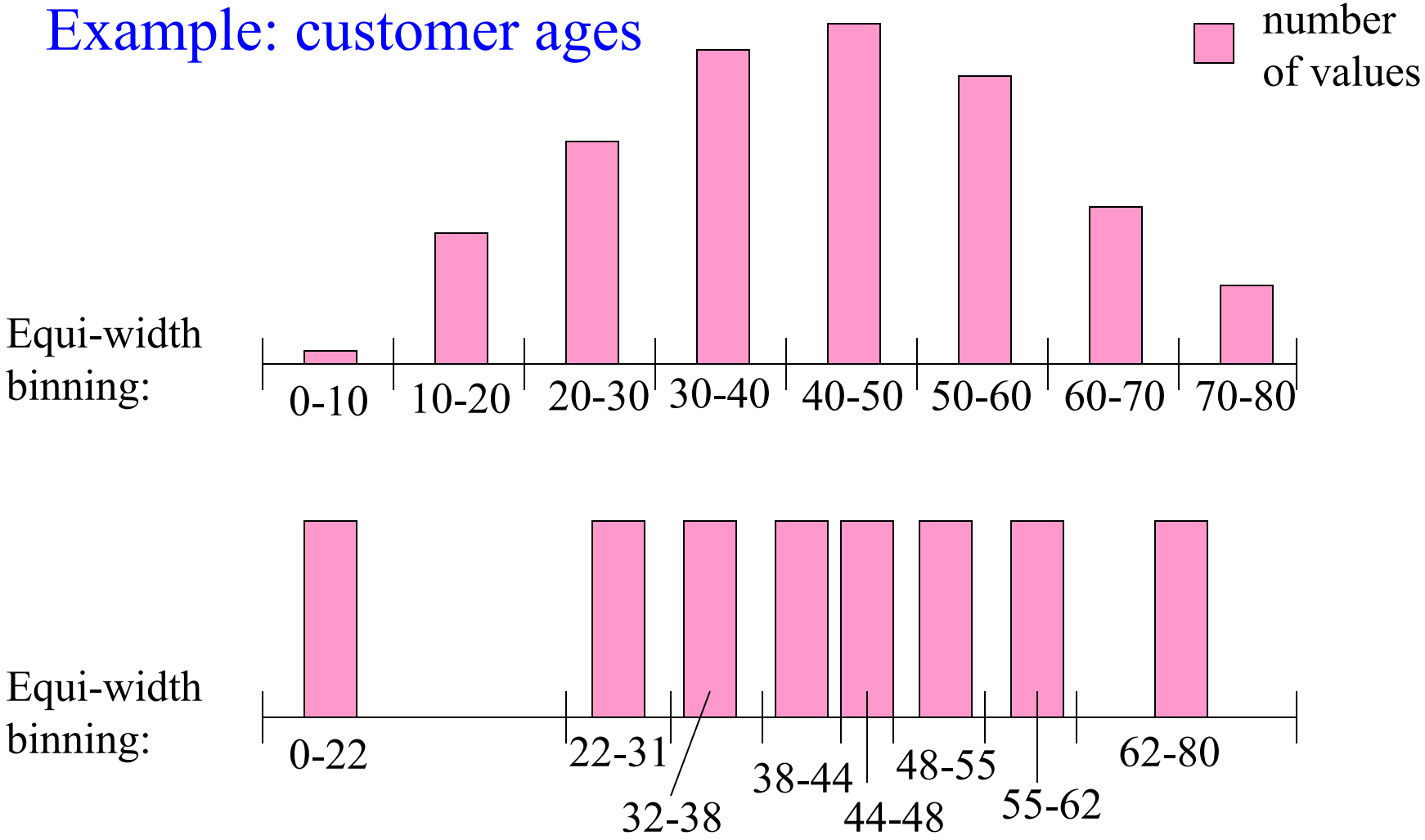
- Binning method:
  - first sort data and partition into (equi-depth) bins
  - then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
- Clustering
  - detect and remove outliers
- Combined computer and human inspection
  - computer detects suspicious values, which are then checked by humans
- Regression
  - smooth by fitting the data into regression functions
- Use Concept hierarchies
  - use concept hierarchies, e.g., price value -> “expensive”

Simple Discretization Methods: Binning

- Equal-width (distance) partitioning:
  - It divides the range into *N* intervals of equal size: uniform grid
  - if *A* and *B* are the lowest and highest values of the attribute, the width of intervals will be: $W = (B-A)/N$.
  - The most straightforward
  - But outliers may dominate presentation
  - Skewed data is not handled well.
- Equal-depth (frequency) partitioning:
  - It divides the range into *N* intervals, each containing approximately same number of samples
  - Good data scaling – good handing of skewed data

Simple Discretization Methods: Binning

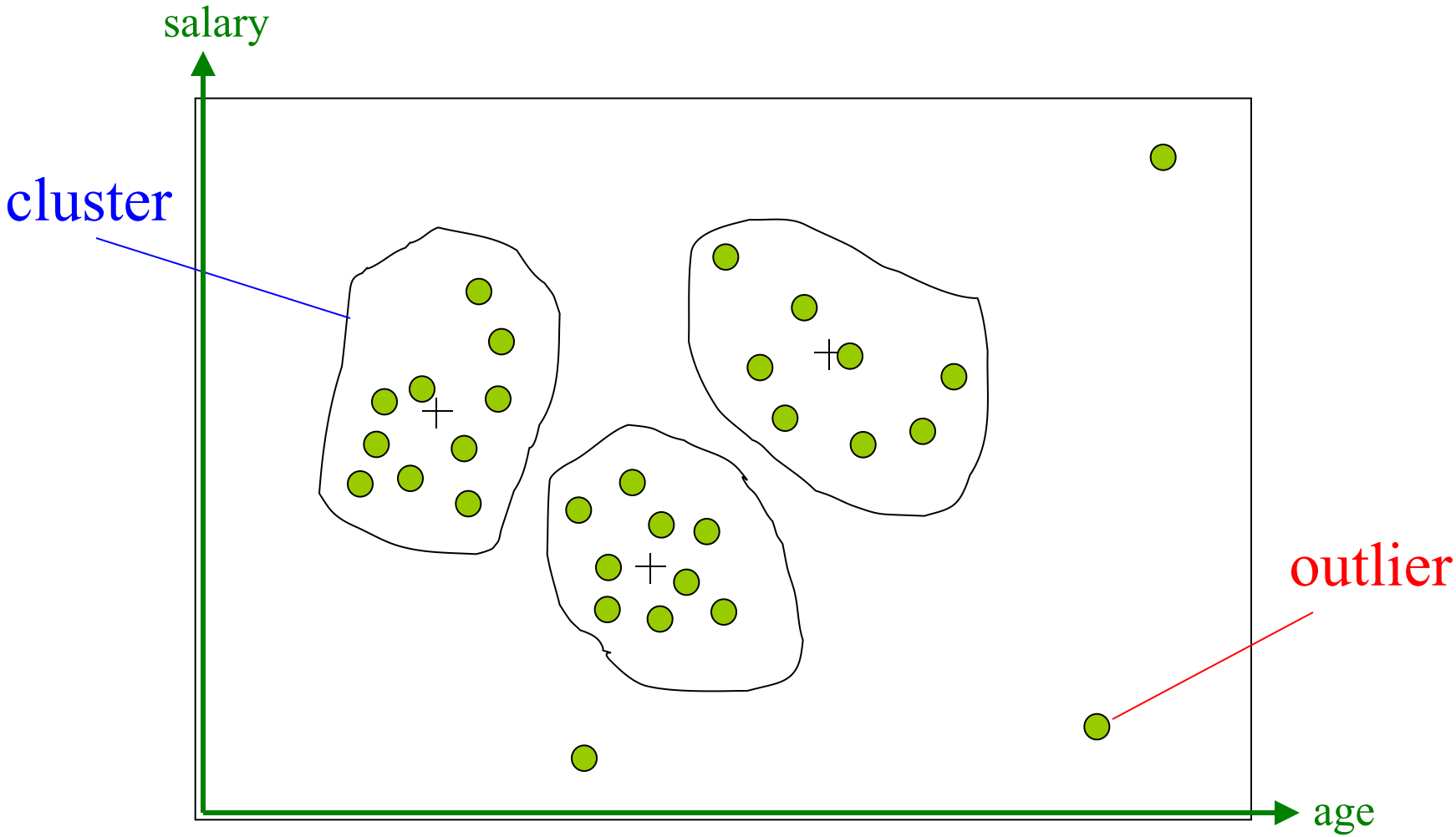
Example: customer ages



Smoothing using Binning Methods

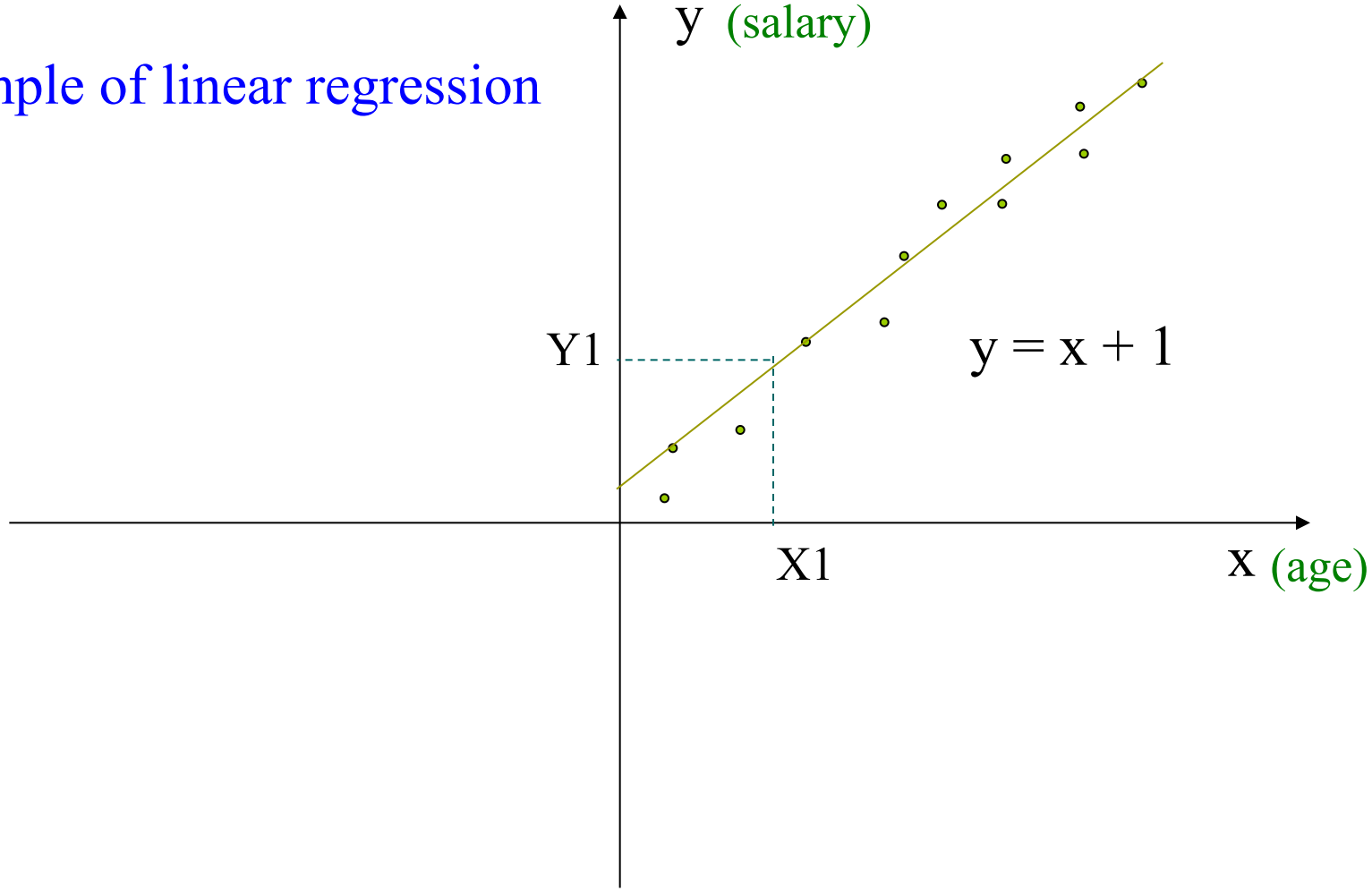
* Sorted data for price (in dollars): 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34
* Partition into (equi-depth) bins:
  - Bin 1: 4, 8, 9, 15
  - Bin 2: 21, 21, 24, 25
  - Bin 3: 26, 28, 29, 34
* Smoothing by bin means:
  - Bin 1: 9, 9, 9, 9
  - Bin 2: 23, 23, 23, 23
  - Bin 3: 29, 29, 29, 29
* Smoothing by bin boundaries: [4,15],[21,25],[26,34]
  - Bin 1: 4, 4, 4, 15
  - Bin 2: 21, 21, 25, 25
  - Bin 3: 26, 26, 26, 34

Cluster Analysis



Regression

Example of linear regression



Inconsistent Data

- Inconsistent data are handled by:
  - Manual correction (expensive and tedious)
  - Use routines designed to detect inconsistencies and manually correct them. E.g., the routine may use the check global constraints (age>10) or functional dependencies
  - Other inconsistencies (e.g., between names of the same attribute) can be corrected during the data integration process

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Data Integration

- Data integration:
  - combines data from multiple sources into a coherent store
- Schema integration
  - integrate metadata from different sources
    - metadata: data about the data (i.e., data descriptors)
  - Entity identification problem: identify real world entities from multiple data sources, e.g., A.cust-id ≡ B.cust-#
- Detecting and resolving data value conflicts
  - for the same real world entity, attribute values from different sources are different (e.g., J.D.Smith and Jonh Smith may refer to the same person)
  - possible reasons: different representations, different scales, e.g., metric vs. British units (inches vs. cm)

Handling Redundant Data in Data Integration

- Redundant data occur often when integration of multiple databases
  - The same attribute may have different names in different databases
  - One attribute may be a "derived" attribute in another table, e.g., annual revenue
- Redundant data may be able to be detected by correlation analysis
- Careful integration of the data from multiple sources may help reduce/avoid redundancies and inconsistencies and improve mining speed and quality

Data Transformation

- Smoothing: remove noise from data
- Aggregation: summarization, data cube construction
- Generalization: concept hierarchy climbing
- Normalization: scaled to fall within a small, specified range
  - min-max normalization
  - z-score normalization
  - normalization by decimal scaling
- Attribute/feature construction
  - New attributes constructed from the given ones

Normalization: Why normalization?

- Speeds-up some learning techniques (ex. neural networks)
- Helps prevent attributes with large ranges outweigh ones with small ranges
  - Example:
    - income has range 3000-200000
    - age has range 10-80
    - gender has domain M/F

Data Transformation: Normalization

- min-max normalization

$$v' = \frac{v - min_A}{max_A - min_A}(new_max_A - new_min_A) + new_min_A$$

  - e.g. convert age=30 to range 0-1, when min=10,max=80. new_age=(30-10)/(80-10)=2/7

- z-score normalization

$$v' = \frac{v - mean_A}{stand_dev_A}$$

- normalization by decimal scaling

$$v' = \frac{v}{10^j}$$ Where j is the smallest integer such that Max($|v'|$)<1

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Data Reduction Strategies

- Warehouse may store terabytes of data: Complex data analysis/mining may take a very long time to run on the complete data set
- Data reduction
  - Obtains a reduced representation of the data set that is much smaller in volume but yet produces the same (or almost the same) analytical results
- Data reduction strategies
  - Data cube aggregation
  - Dimensionality reduction
  - Data compression
  - Numerosity reduction
  - Discretization and concept hierarchy generation

Data Cube Aggregation

- The lowest level of a data cube
  - the aggregated data for an individual entity of interest
  - e.g., a customer in a phone calling data warehouse.
- Multiple levels of aggregation in data cubes
  - Further reduce the size of data to deal with
- Reference appropriate levels
  - Use the smallest representation which is enough to solve the task
- Queries regarding aggregated information should be answered using data cube, when possible

Dimensionality Reduction

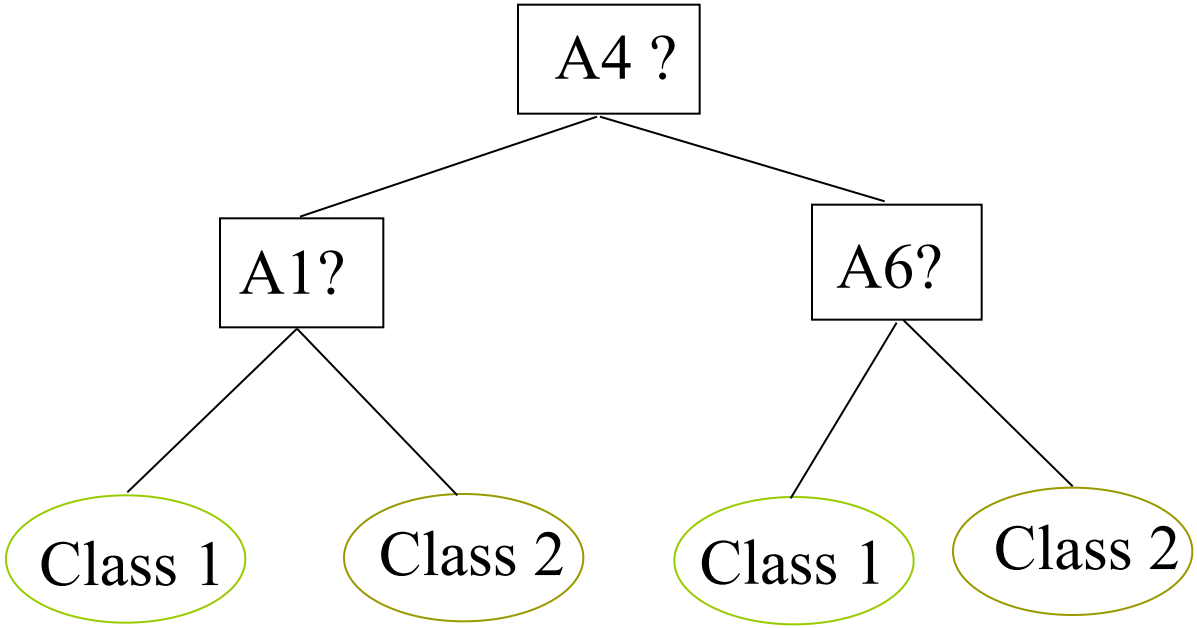
- Feature selection (i.e., attribute subset selection):
  - Select a minimum set of features such that the probability distribution of different classes given the values for those features is as close as possible to the original distribution given the values of all features
  - reduce # of patterns in the patterns, easier to understand
- Heuristic methods (due to exponential # of choices):
  - step-wise forward selection
  - step-wise backward elimination
  - combining forward selection and backward elimination
  - decision-tree induction

Heuristic Feature Selection Methods

- There are 2^d possible sub-features of d features
- Several heuristic feature selection methods:
  - Best single features under the feature independence assumption: choose by significance tests.
  - Best step-wise feature selection:
    - The best single-feature is picked first
    - Then next best feature condition to the first, ...
  - Step-wise feature elimination:
    - Repeatedly eliminate the worst feature
  - Best combined feature selection and elimination:
  - Optimal branch and bound:
    - Use feature elimination and backtracking

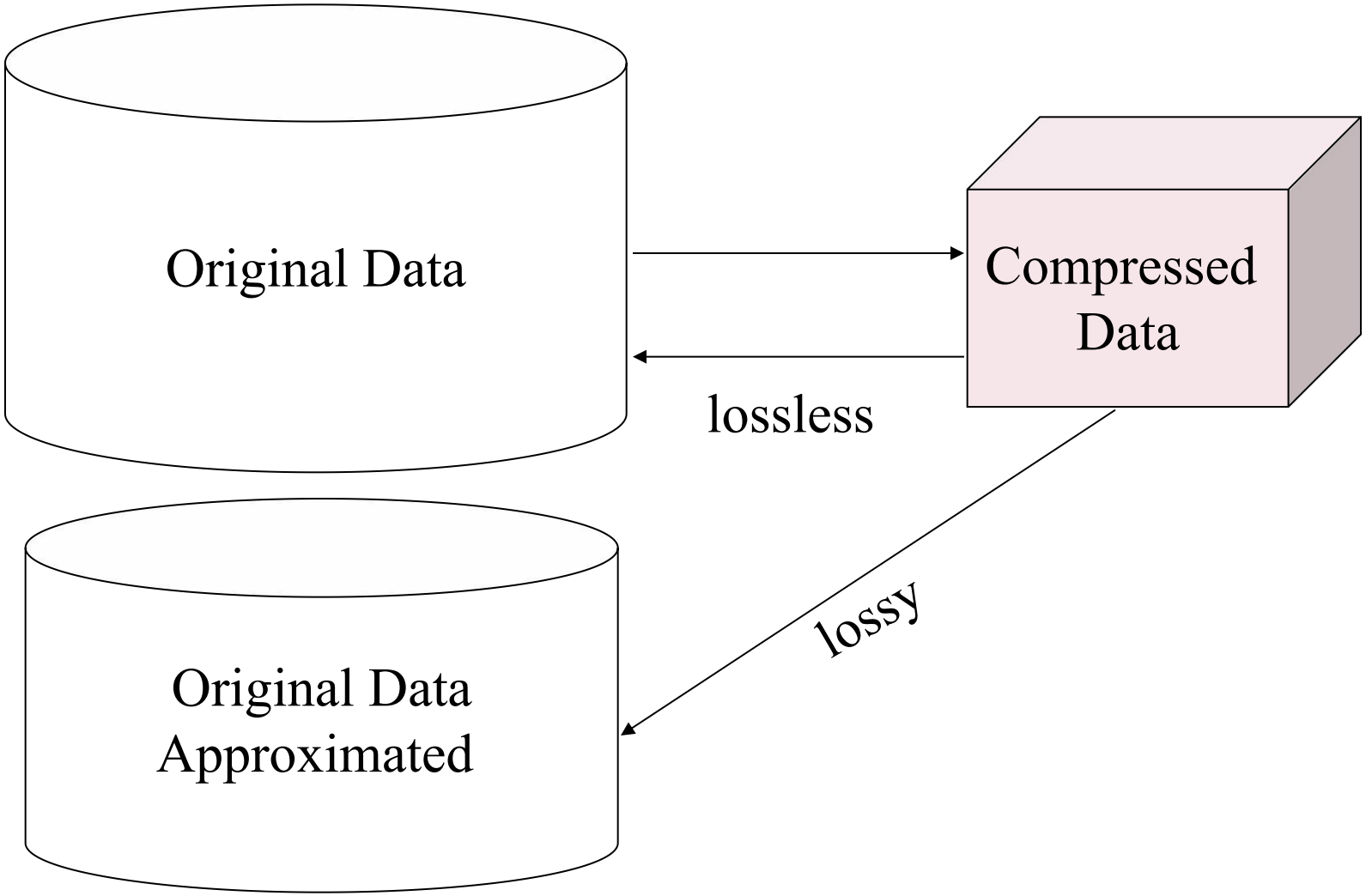
Example of Decision Tree Induction

Initial attribute set:
{A1, A2, A3, A4, A5, A6}



-------> Reduced attribute set: {A1, A4, A6}

Data Compression



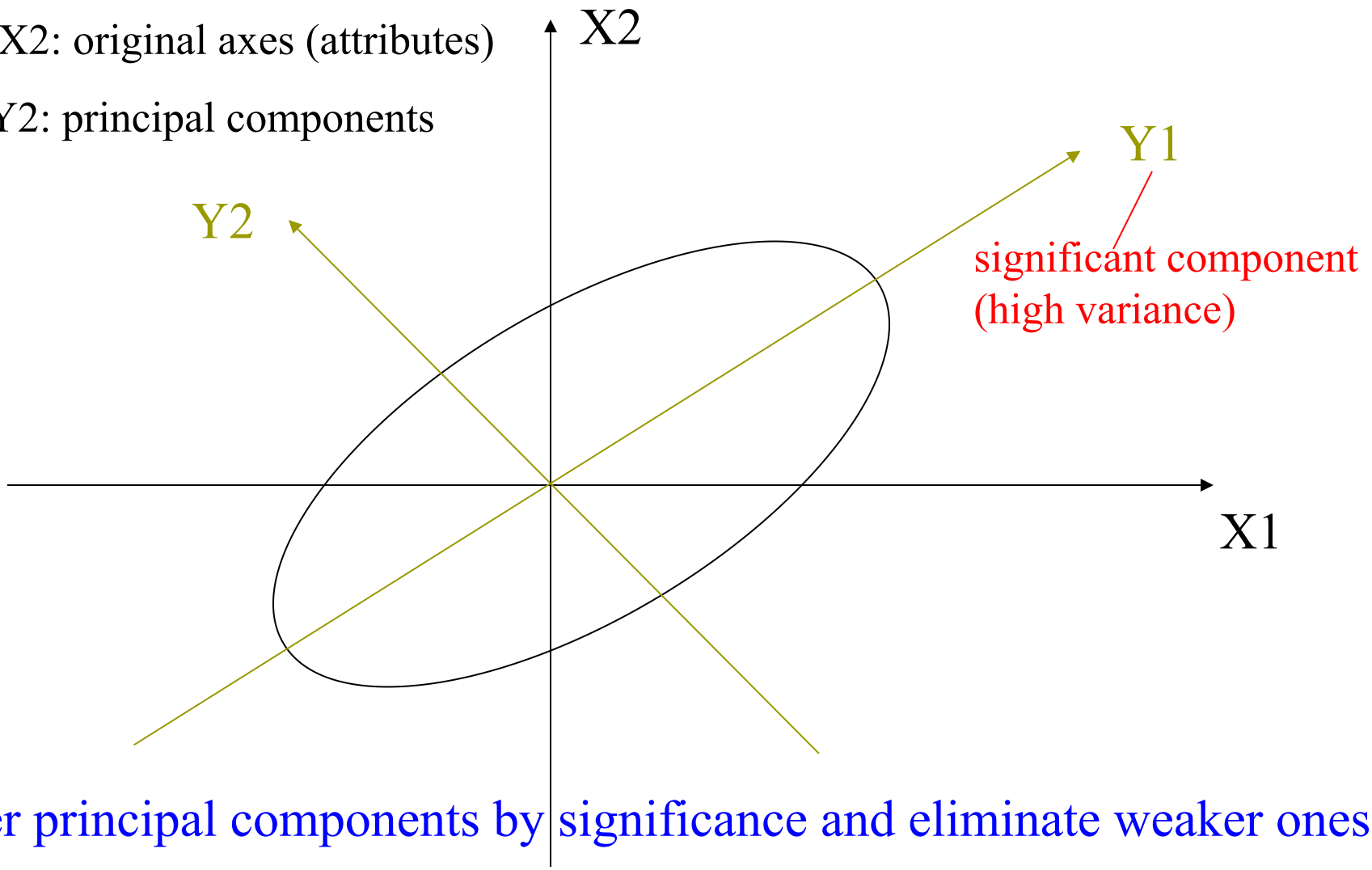
Principal Component Analysis or Karhuren-Loeve (K-L) method

- Given *N* data vectors from *k*-dimensions, find *c* <= *k* orthogonal vectors that can be best used to represent data
  - The original data set is reduced to one consisting of N data vectors on c principal components (reduced dimensions)
- Each data vector is a linear combination of the *c* principal component vectors
- Works for numeric data only
- Used when the number of dimensions is large

Principal Component Analysis

X1, X2: original axes (attributes)

Y1,Y2: principal components



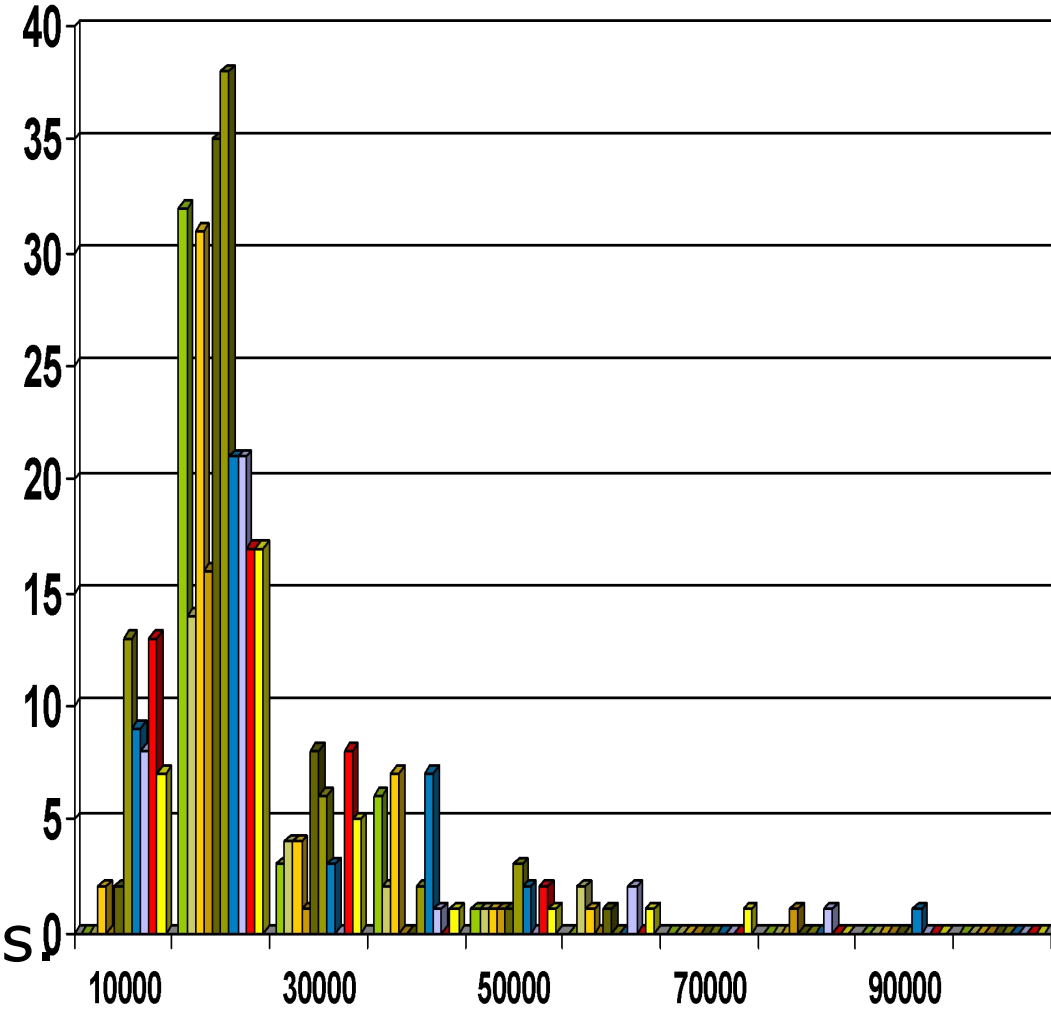
Order principal components by significance and eliminate weaker ones

Numerosity Reduction: Reduce the **volume** of data

- Parametric methods
  - Assume the data fits some model, estimate model parameters, store only the parameters, and discard the data (except possible outliers)
  - Log-linear models: obtain value at a point in m-D space as the product on appropriate marginal subspaces
- Non-parametric methods
  - Do not assume models
  - Major families: histograms, clustering, sampling

Histograms

- A popular data reduction technique
- Divide data into buckets and store average (or sum) for each bucket
- Can be constructed optimally in one dimension using dynamic programming
- Related to quantization problems



Histogram types

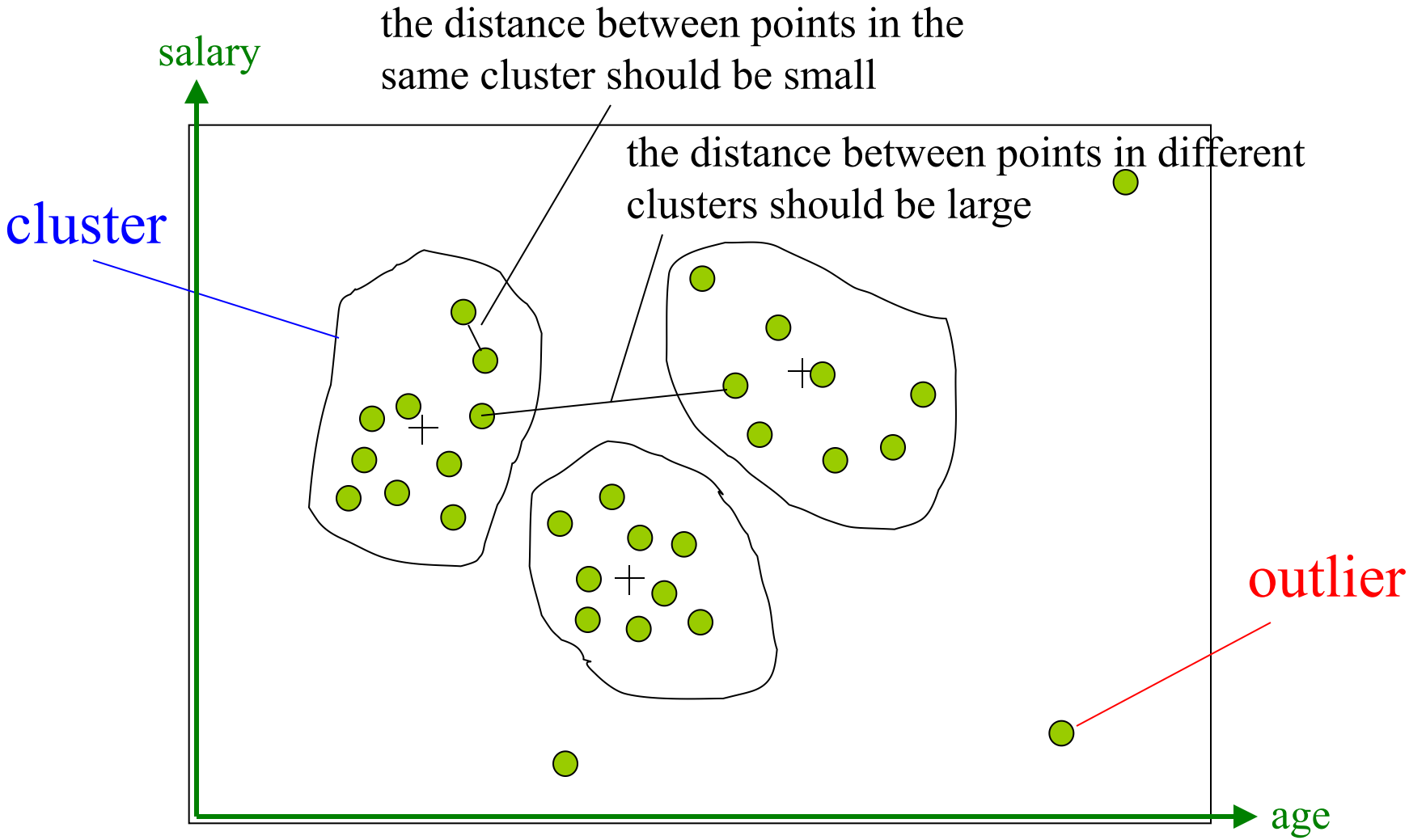
- Equal-width histograms:
  - It divides the range into *N* intervals of equal size
- Equal-depth (frequency) partitioning:
  - It divides the range into *N* intervals, each containing approximately same number of samples
- V-optimal:
  - It considers all histogram types for a given number of buckets and chooses the one with the least variance.
- MaxDiff:
  - After sorting the data to be approximated, it defines the borders of the buckets at points where the adjacent values have the maximum difference
    - Example: split 1,1,4,5,5,7,9,14,16,18,27,30,30,32 to three buckets

MaxDiff 27-18 and 14-9

Clustering

- Partitions data set into clusters, and models it by one representative from each cluster
- Can be very effective if data is clustered but not if data is “smeared”
- There are many choices of clustering definitions and clustering algorithms, further detailed in Chapter 7

Cluster Analysis



Hierarchical Reduction

- Use multi-resolution structure with different degrees of reduction
- Hierarchical clustering is often performed but tends to define partitions of data sets rather than "clusters"
- Parametric methods are usually not amenable to hierarchical representation
- Hierarchical aggregation
  - An index tree hierarchically divides a data set into partitions by value range of some attributes
  - Each partition can be considered as a bucket
  - Thus an index tree with aggregates stored at each node is a hierarchical histogram

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Discretization

- Three types of attributes:
  - Nominal — values from an unordered set
  - Ordinal — values from an ordered set
  - Continuous — real numbers
- Discretization:
  - divide the range of a continuous attribute into intervals
  - why?
    - Some classification algorithms only accept categorical attributes.
    - Reduce data size by discretization
    - Prepare for further analysis

Discretization and Concept hierachy

- Discretization
  - reduce the number of values for a given continuous attribute by dividing the range of the attribute into intervals. Interval labels can then be used to replace actual data values.
- Concept hierarchies
  - reduce the data by collecting and replacing low level concepts (such as numeric values for the attribute age) by higher level concepts (such as young, middle-aged, or senior).

Discretization and concept hierarchy generation for numeric data

- Binning/Smoothing (see sections before)
- Histogram analysis (see sections before)
- Clustering analysis (see sections before)
- Entropy-based discretization
- Segmentation by natural partitioning

Entropy: $Ent(S_1) = -\sum_{i=1}^{m} p_i \log_2(p_i)$

Entropy-Based Discretization

- Given a set of samples S, if S is partitioned into two intervals S1 and S2 using boundary T, the information gain I(S,T) after partitioning is

$$I(S,T) = \frac{|S_1|}{|S|} Ent(S_1) + \frac{|S_2|}{|S|} Ent(S_2)$$

- The boundary that maximizes the information gain over all possible boundaries is selected as a binary discretization.
- The process is recursively applied to partitions obtained until some stopping criterion is met, e.g.,

$$Ent(S) - I(T,S) > \delta$$

- Experiments show that it may reduce data size and improve classification accuracy

Segmentation by natural partitioning

- Users often like to see numerical ranges partitioned into relatively uniform, easy-to-read intervals that appear intuitive or "natural". E.g., [50-60] better than [51.223-60.812]

The 3-4-5 rule can be used to segment numerical data into relatively uniform, "natural" intervals.

* If an interval covers 3, 6, 7 or 9 distinct values at the most significant digit, partition the range into 3 equiwidth intervals for 3,6,9 or 2-3-2 for 7
* If it covers 2, 4, or 8 distinct values at the most significant digit, partition the range into 4 equiwidth intervals
* If it covers 1, 5, or 10 distinct values at the most significant digit, partition the range into 5 equiwidth intervals

The rule can be recursively applied for the resulting intervals

Concept hierarchy generation for categorical data

- Categorical attributes: finite, possibly large domain, with no ordering among the values
  - Example: item type
- Specification of a partial ordering of attributes explicitly at the schema level by users or experts
  - Example: location is split by domain experts to street<city<state<country
- Specification of a portion of a hierarchy by explicit data grouping
- Specification of a set of attributes, but not of their partial ordering
- Specification of only a partial set of attributes

Specification of a set of attributes

Concept hierarchy can be automatically generated based on the number of distinct values per attribute in the given attribute set. The attribute with the most distinct values is placed at the lowest level of the hierarchy.

